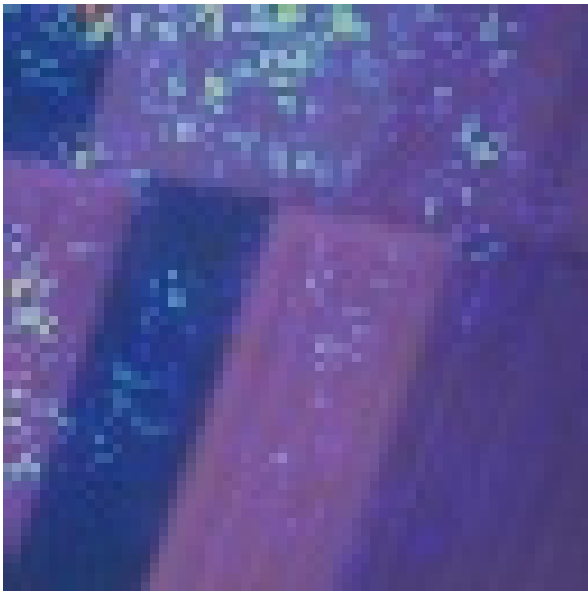


Satellite Land Classification Report

Record ID:	134	Timestamp:	2026-06-30 08:19:09
Image Name:	AnnualCrop_30.jpg	Confidence:	100.00%
Classification:	AnnualCrop	Model:	MobileNetV2 (Transfer Learning)



Original Satellite Image



Grad-CAM Explainability Heatmap

Land Cover Description

Area designated for crops that are planted and harvested annually (e.g., wheat, maize). These regions are vital for food security but require careful agricultural management to prevent soil degradation.

Grad-CAM Explainability Analysis

The Grad-CAM heatmap highlights the specific pixel regions that the deep learning model (MobileNetV2) focused on to determine the class. Red and yellow regions indicate high-intensity features (like edges, buildings, or water textures) that heavily influenced the classification, while blue regions represents elements that had negligible impact. This validation ensures that the model is identifying genuine geographical features rather than relying on peripheral noise.

This report is automatically generated by the Satellite Land Classification and Change Detection System dashboard. Classification accuracy is based on training validations on the EuroSAT dataset.